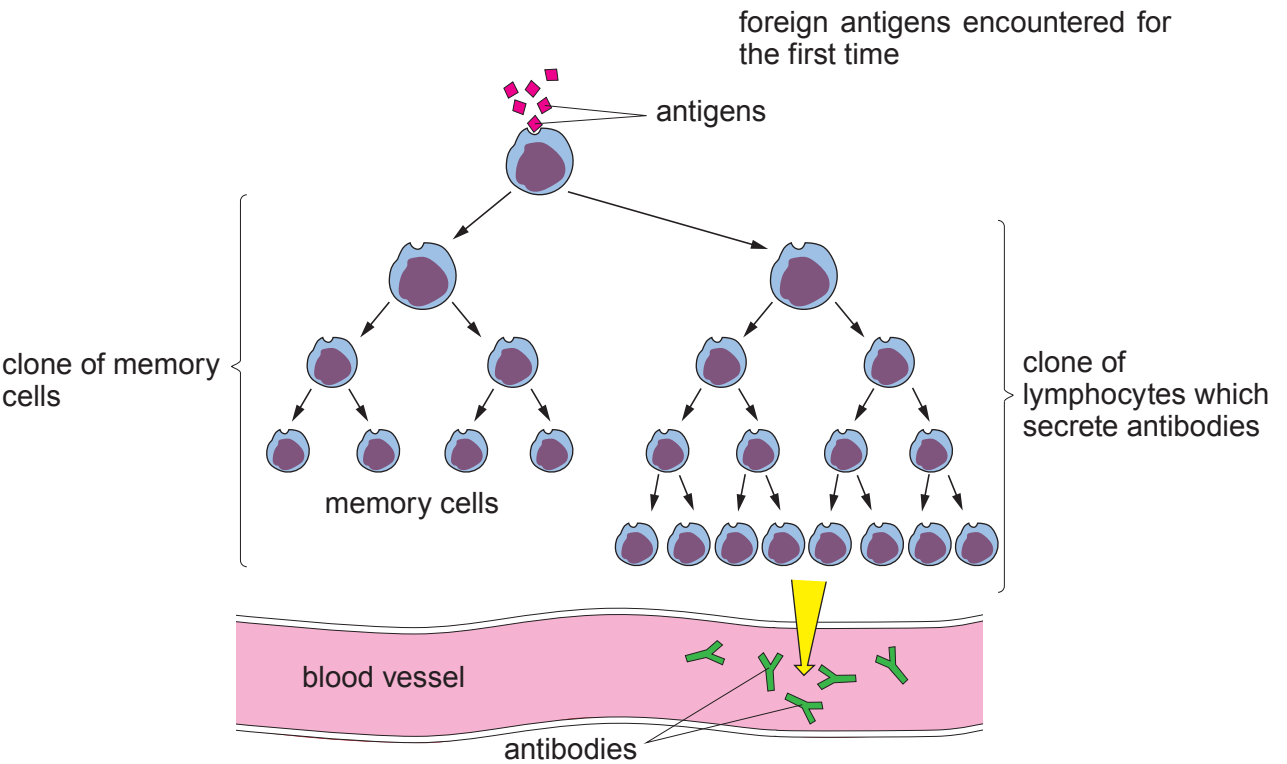


7. The diagram below shows how a lymphocyte responds to the presence of antigens in the body.



(a) Explain how the body responds to a foreign antigen the first time it is encountered and why the response is faster when the same antigen is encountered a second time. [3]

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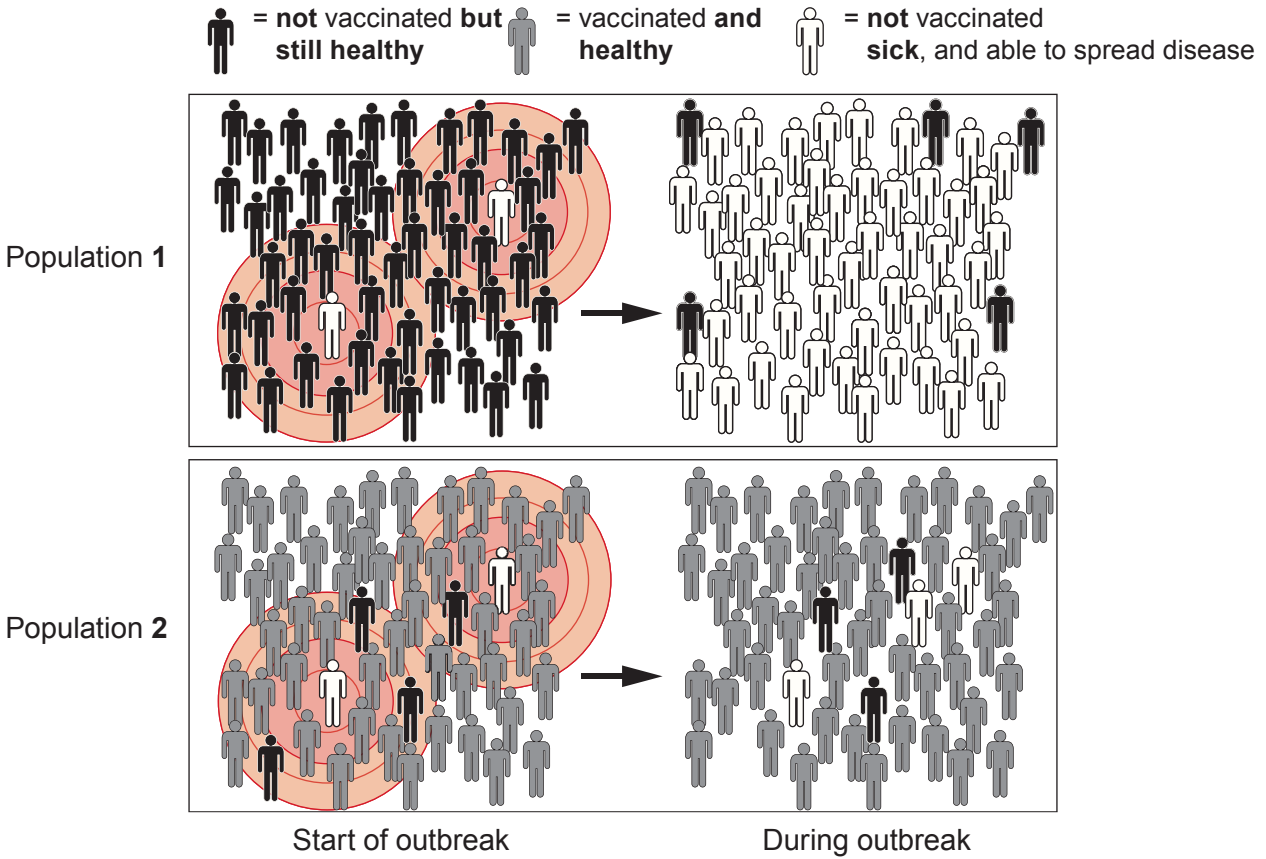
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(b) Measles is a communicable disease. The diagram shows two populations of people, during an outbreak of the measles virus.



Using **only the diagram above**, explain the advantage of a vaccination programme. [2]

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